Vulvovaginal Candidiasis (VVC)

Divine Aleru

Microbiome Signatures, Microbiome Medicine

Abstract. VVC is caused by Candida albicans overgrowth. It is influenced by microbiome imbalances, immune factors, and requires effective treatments.

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Overview

Vulvovaginal candidiasis (VVC) is a highly prevalent fungal infection predominantly caused by *Candida albicans* and occasionally by non-albicans species such as *C. glabrata* and *C. krusei*. It affects approximately 70-75% of women at least once in their lifetime, with 5-8% experiencing recurrent vulvovaginal candidosis (RVVC), defined as four or more symptomatic episodes per year.[1] The disease manifests as an acute inflammatory condition of the vulva and vaginal mucosa, characterized by itching, burning, redness, and abnormal vaginal discharge. The pathogenesis is multifactorial, involving complex interactions among the Candida species' virulence factors, host immune responses, estrogenic environment, and vaginal microbiome, primarily Lactobacilli.[2] Candida's ability to switch between yeast and hyphal forms, form biofilms, and secrete toxins such as candidalysin contributes to infection persistence and symptom severity. Diagnosis remains challenging due to the presence of Candida as a commensal in many asymptomatic women.[3] Treatment mainly relies on azole antifungals, though increasing azole resistance, especially among non-albicans species, complicates management.[4][5] Long-term maintenance antifungal regimens are often necessary for RVVC but are associated with risks of resistance and relapse. Patient quality of life is significantly affected due to physical discomfort, psychological burden, and impaired sexual health. Advancing diagnostic accuracy, understanding microbiome-host interactions, and developing novel therapeutics are critical needs for improving clinical outcomes.

Associated Conditions

RVVC is linked with several associated conditions that modulate susceptibility and severity. High estrogen states such as pregnancy, use of oral contraceptives, and hormone replacement therapy increase the risk by altering vaginal immunity and microbiome balance.[6] Diabetes mellitus and immunosuppressive states, including HIV infection, also predispose women to frequent recurrences.[7][8] Antibiotic usage disrupts the protective Lactobacillus-dominant vaginal microbiota, facilitating Candida overgrowth.[9] Behavioral factors such as frequent sexual intercourse, use of poorly ventilated clothing, and atopic diseases may contribute to susceptibility. Genetic polymorphisms affecting innate immune signaling pathways have been

implicated in impaired fungal clearance and enhanced inflammation. These factors, individually or in combination, create a permissive environment for Candida pathogenicity and chronic infection.

Causal Theories

The etiology of VVC remains incompletely understood and is recognized as multifactorial. The primary causal theory revolves around an imbalance between Candida virulence mechanisms and host immune defenses. Candida albicans employs a polymorphic lifestyle, switching from benign yeast to invasive hyphal forms, secreting enzymes and candidalysin that promote tissue invasion and inflammation. Biofilm formation further shields Candida from antifungal agents.[10] Host factors impair fungal clearance, including estrogen-driven changes, neutrophil dysfunction, and genetic variants in pattern recognition receptors. The vaginal microbiome, particularly *Lactobacilli*, usually suppresses Candida growth; disruptions to this ecosystem via antibiotics or other factors remove this inhibition.[11] Limitations in current causal theories include insufficient understanding of the triggers converting asymptomatic colonization into symptomatic infection and the role of non-cultivable fungal species. Novel hypotheses suggest that immunological hypersensitivity to Candida antigens may drive symptoms independent of fungal load, highlighting the complexity of RVVC pathogenesis.[12]

Theory	Description
Candida Virulence & Morphogenesis	Candida's morphological switching and virulence factor secretion lead to tissue damage
Host Immune Dysfunction	Genetic and functional immune defects impair fungal clearance
Microbiome Dysbiosis	Disruption of Lactobacillus-dominated vaginal flora facilitates Candida overgrowth
Hypersensitivity Response	Exaggerated inflammatory reaction to Candida antigens causes symptoms

Diagnosis

Accurate diagnosis of VVC requires confirmation of Candida presence combined with clinical symptoms. Conventional diagnosis relies on clinical examination, microscopy to detect pseudohyphae or blastospores, and culture to identify Candida species.[13] However, microscopy sensitivity ranges from 50-80%, and culture may be slow or inconclusive. Molecular methods such as PCR enhance detection sensitivity and can identify mixed or resistant strains, but are not widely accessible.[14] Identification of Candida species is crucial, as non-albicans species often exhibit reduced susceptibility to azoles. Antifungal susceptibility testing is recommended in refractory cases.[15] Emerging research into microbiome and metabolomic signatures holds promise for non-invasive diagnostics that distinguish colonization from infection and identify treatment-resistant profiles.[16] Nonetheless, lack of point-of-care rapid testing and inconsistent guideline adherence contribute to delayed or inappropriate treatment, underscoring the need for improved diagnostics integrating microbiome insights.

Primer

VVC cannot be fully understood without considering its relationship with the vaginal microbiome, where beneficial bacteria such as *Lactobacillus* species maintain an acidic environment that suppresses fungal overgrowth. Beyond microbial interactions, the vaginal environment's metal content plays a vital role. Essential metals like zinc and copper are tightly controlled by the host through nutritional immunity mechanisms to restrict fungal growth. Meanwhile, *Candida* has evolved strategies to acquire these metals despite host defenses. This dynamic interplay between the host's immune system, resident microbes, and metal availability forms a complex network that shapes the course of VVC and offers new directions for diagnosis and treatment.

Metallomic Signatures

The metallomic signature of VVC reflects how *Candida albicans* interacts with and adapts to metal stress in the host environment. Research shows that *C. albicans* can tolerate and even sequester high concentrations of toxic metals like chromium, lead, and zinc, using biosorption mechanisms involving its cell wall.[17][x] These adaptations not only enhance its survival in hostile environments but may also reinforce its biofilm structure and resistance to antifungal agents. Additionally, differential sensitivity to metals like cadmium and mercury indicates the presence of complex detoxification and efflux pathways, which may overlap with drug resistance mechanisms. From a therapeutic perspective, disrupting *Candida's* metal homeostasis, or exploiting its biosorptive capacity, could offer novel treatment strategies, particularly in drug-resistant or recurrent VVC cases. This metallomic insight expands our understanding of the infection beyond microbiome shifts, adding a crucial chemical-ecological layer to host-pathogen interactions.

▶ **What is the Metallomic Signature of Vulvovaginal candidiasis?**

Nutritional Immunity

Nutritional immunity is a key host defense mechanism in VVC, where the body restricts access to essential metals like zinc and copper to limit the growth and virulence of *Candida albicans*. During infection, immune cells such as neutrophils release calprotectin, a protein that tightly binds these metals, effectively starving the fungus of nutrients it needs to survive.[26] In response, *C. albicans* activates specialized metal-scavenging systems, including the zinc-binding protein PRA1 and transporter ZRT1, and shifts its antioxidant defenses to rely on manganese when copper is unavailable.[27] This ongoing struggle between host and pathogen, essentially a biochemical tug-of-war over micronutrients, plays a critical role in infection severity and persistence. When the vaginal microbiota, particularly protective *Lactobacillus* species, is disrupted, this defense can weaken, giving *Candida* a competitive edge.[28] Understanding nutritional immunity not only sheds light on the pathogenesis of VVC but also suggests new therapeutic directions, including strategies that enhance host metal-binding defenses or target fungal nutrient acquisition systems.

Interventions

Our validation method evaluates the efficacy of microbiome-targeted interventions for VVC by correlating their influence on vaginal microbial balance, clinical symptom resolution, and the modulation of pathogenic mechanisms such as *Candida* overgrowth and biofilm formation. This approach distinguishes interventions that are validated, promising, or experimental based on the strength of evidence connecting changes in the vaginal microbiome with clinical improvement and relevant microbiological signatures.

Intervention	Classification	Mechanism of Action	MBTI Status
Probiotics	Supplement	Replenish beneficial vaginal microbiota, competitively inhibit *Candida*, acidify vaginal pH, produce antifungal compounds and immunomodulate host response.[29] [30]	Validated
Postbiotics	Supplement	Bioactive compounds released by probiotics, such as bacteriocins, SCFAs, and enzymes that inhibit pathogen growth and modulate inflammation.[31][x]	Promising Candidate
Prebiotics/Synbiotics	Supplement	Selectively enhance growth of beneficial microbes, thus indirectly suppressing *Candida* overgrowth; synbiotics combine prebiotics and probiotics.[32][33][x]	Promising Candidate
Vaginal Microbiome Transplantation (VMT)	Microbiota-based therapy	Transplant healthy donor vaginal microbiota to restore eubiosis and *Lactobacillus* dominance, reversing dysbiosis linked to VVC.[34][x]	Promising Candidate
Short chain fatty acid supplements	Supplement	Modulate vaginal pH, inhibit *Candida* growth and virulence, support mucosal immunity.[35][x][36]	Under investigation
Lactoferrin	Supplement	Chelates iron and other metals, impairs *Candida* growth, modulates inflammation, supports mucosal immunity.[37][38]	Promising candidate
Dietary Supplementation to maintain optimal calprotectin responses	Diet	Calciprotein binds and withholds essential metals (Zn, Cu), induces nutritional immunity, inhibits *Candida* growth and virulence gene expression.[39]	Experimental
Boric acid	Pharmaceutical	Disrupts *Candida* cell wall integrity, inhibits biofilm formation, restores vaginal acidity, effective against non-albicans and azole-resistant *Candida* species.[40]	Validated

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